

SMART PARKING PLATFORM

Project Management Plan
WE ARE \$oftware ¢orp.
Version 0.3 – Homework 1–3
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1. Vision and Scope

WE ARE \$oftware \$orp. is developing the Smart Parking Platform as an internal commercial product. The web and mobile platform will help drivers locate available parking, reserve spaces, pay digitally, and navigate to participating garages. Parking operators will manage facilities, pricing, availability, reservations, and reporting.

1.1 Scope

In Scope	Out of Scope
Registration/authentication; real-time availability; map/search; facility details; reservations; payments; history/receipts; operator garage/pricing/inventory management; reports; notifications; mapping/navigation integration.	Physical parking hardware; garage construction/operation; parking enforcement/towing; proprietary GPS infrastructure; physical equipment maintenance.

1.2 Stakeholders and Constraints

Stakeholders	Key Constraints
Drivers; parking operators; city transportation departments; system administrators; finance/billing personnel; mobile users; third-party payment providers.	Fixed semester timeline; third-party API/service dependencies; security/privacy requirements; city ordinances and laws; unlimited theoretical development budget.

2. Competitive Analysis

Platform	Strengths / Existing Capabilities	Project Differentiation Opportunity
SpotHero	Parking search, reservations, prepaid parking, and operator tools.	Create a consistent workflow connecting live availability, reservations, payments, navigation, and operator updates.
ParkWhiz	Parking reservations, payments, and operator reporting/integrations.	Emphasize standardized facility data and a consistent experience across participating garages.
ParkMobile	Reservations plus parking-management, pricing, analytics, and provider tools.	Connect driver and operator workflows through the same real-time parking data.

3. Software Requirements Specification

3.1 Functional Requirements

ID	Requirement
FR-01	Register and authenticate users; support password reset and logout.
FR-02	Search for parking by location, destination, date, or time.
FR-03	Display parking locations on an interactive map.
FR-04	Display facility details, pricing, restrictions, and availability.
FR-05	Display real-time parking availability from supported data sources.
FR-06	Create and cancel parking reservations.
FR-07	Process digital payments through an approved payment provider.
FR-08	Display reservation history, confirmations, and receipts.
FR-09	Send reservation and account notifications.
FR-10	Launch supported mapping/navigation for a selected facility.

FR-11	Allow operators to add, edit, and deactivate facilities.
FR-12	Allow operators to manage pricing, inventory, and availability.
FR-13	Allow operators to review reservations and facility activity.
FR-14	Provide occupancy and utilization reporting.
FR-15	Provide authorized financial and transaction reporting/export.

3.2 Non-Functional Requirements

ID	Requirement
NFR-01	Encrypted communications and secure password storage.
NFR-02	Role-based access control.
NFR-03	Target 99.9% availability, excluding planned maintenance and third-party outages.
NFR-04	Common requests should normally respond within 3 seconds, excluding third-party delays.
NFR-05	Responsive web/mobile interface.
NFR-06	Scalable architecture for growth in users, facilities, and transactions.
NFR-07	Audit logging for important administrative and financial actions.
NFR-08	Graceful handling of third-party service failures.

4. Use Cases

Use Case	Actor	Main Flow
UC-01 Register Account	Driver / Operator	Enter account information → validate → create account.
UC-02 Log In	Driver / Operator / Admin	Enter credentials → validate → open authorized dashboard.
UC-03 Reset Password	Registered User	Request reset → verify identity → create new password.
UC-04 Search Parking	Driver	Enter destination/date/time → display matching facilities.
UC-05 View Parking Map	Driver	Open map → display matching facilities → select facility.
UC-06 View Facility Details	Driver	Select facility → display rates, restrictions, details, and availability.
UC-07 Create Reservation	Driver	Select facility/time → validate inventory → create reservation.
UC-08 Pay for Reservation	Driver	Choose payment → authorize transaction → record payment and receipt.
UC-09 Cancel Reservation	Driver	Select eligible reservation → confirm cancellation → update inventory/status.
UC-10 View History / Receipt	Driver	Open history → select reservation/transaction → display details.
UC-11 Navigate to Facility	Driver	Select directions → send destination to mapping/navigation service.
UC-12 Add Parking Facility	Operator	Enter facility details → validate → save facility.
UC-13 Edit Facility / Pricing	Operator	Edit facility/rates/restrictions → validate → save changes.
UC-14 Manage Availability	Operator	Update parking inventory → validate → refresh availability.
UC-15 Review Reports	Operator / Finance	Choose report/date range → aggregate data → display/export report.

5. Work Breakdown Structure

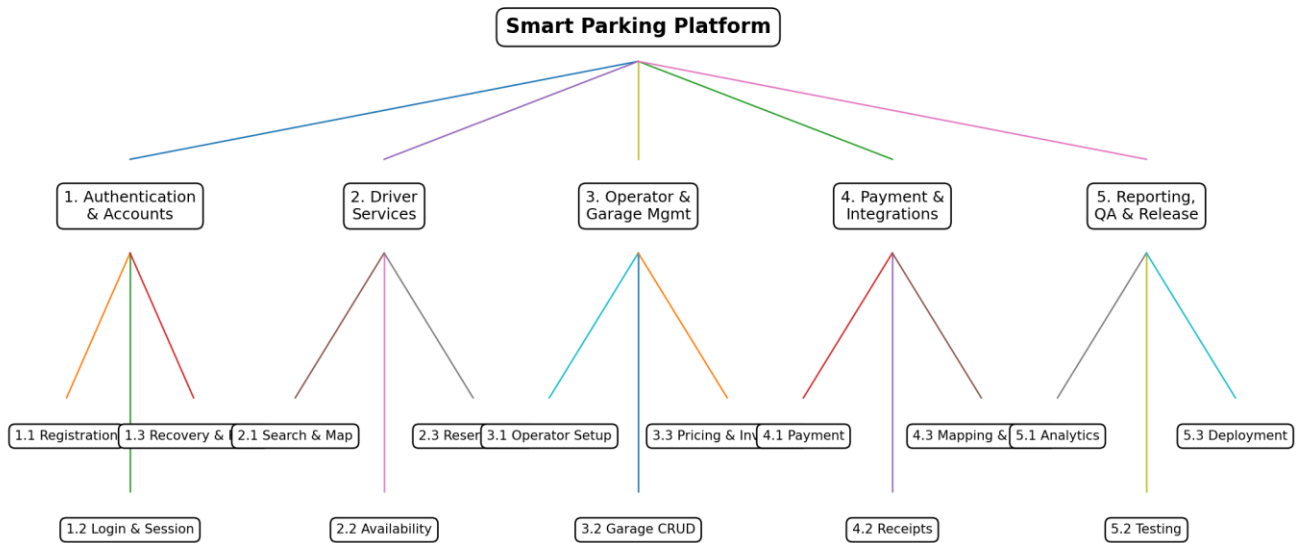


Figure 1. Smart Parking Platform Work Breakdown Structure (three levels or more).

6. Project Schedule

Deliverable	Planned Dates
Requirements Baseline	Sep 1–11
Architecture & UI Design	Sep 8–18
Authentication & Accounts	Sep 14–25
Operator / Garage Management	Sep 21–Oct 9
Driver Search, Map & Availability	Sep 28–Oct 16
Reservations & Payments	Oct 12–30
Reporting & Notifications	Oct 26–Nov 6
System Integration	Nov 2–13
QA & Security Testing	Nov 9–27
Final Release & Documentation	Nov 23–Dec 4

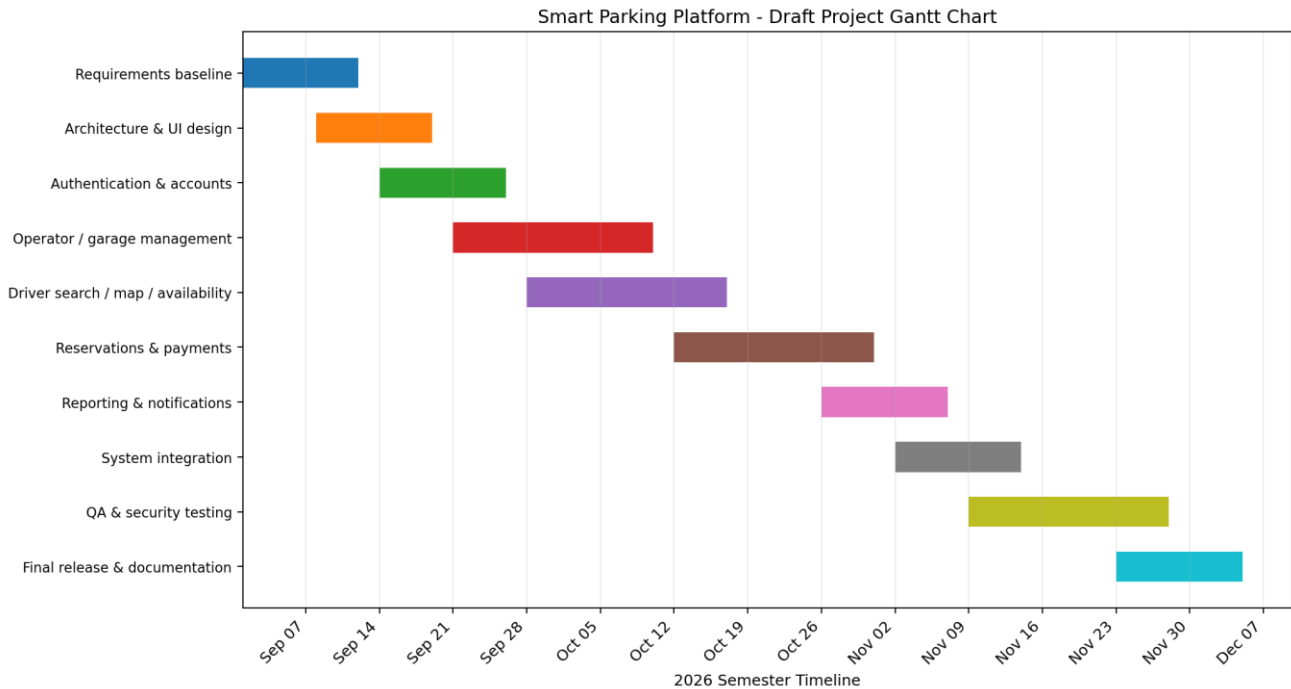


Figure 2. Draft semester Gantt chart.

7. Product Backlog and Sprint 1

Product backlog: 45 items total – 5 Login, 15 UI, 15 Backend, and 10 Reporting.

Category	Backlog Items
Login	LOG-01 Driver Login; LOG-02 Operator Login; LOG-03 User Registration; LOG-04 Password Reset; LOG-05 Session Logout
UI	UI-01 Driver Dashboard; UI-02 Operator Dashboard; UI-03 Parking Search Interface; UI-04 Interactive Parking Map; UI-05 Parking Facility Details; UI-06 Real-Time Availability Display; UI-07 Reservation Screen; UI-08 Payment Checkout Screen; UI-09 Reservation Confirmation Screen; UI-10 Reservation History Screen; UI-11 Receipt View; UI-12 Driver Profile Settings; UI-13 Garage Management Interface; UI-14 Pricing Management Interface; UI-15 Operator Reporting Dashboard
Backend	BE-01 Driver Account Creation API; BE-02 Operator Account Creation API; BE-03 Authentication API; BE-04 Password Recovery Process; BE-05 Parking Facility Search API; BE-06 Real-Time Availability Service; BE-07 Garage Creation Process; BE-08 Garage Update Process; BE-09 Parking Inventory Management; BE-10 Reservation Creation Process; BE-11 Reservation Cancellation Process; BE-12 Payment Processing Integration; BE-13 Receipt Generation Process; BE-14 Mapping & Navigation Integration; BE-15 Notification Service
Reporting	REP-01 Current Occupancy Report; REP-02 Daily Occupancy Report; REP-03 Weekly Occupancy Report; REP-04 Reservation Activity Report; REP-05 Revenue Summary Report; REP-06 Transaction History Report; REP-07 Garage Utilization Report; REP-08 Peak Parking Hours Report; REP-09 Driver Reservation History Report; REP-10 Financial Report Export

7.1 Sprint 1 Backlog

ID	Sprint 1 Item
LOG-02	Operator Login
LOG-03	User Registration
BE-02	Operator Account Creation API
BE-03	Authentication API
BE-06	Real-Time Availability Service
BE-07	Garage Creation Process
BE-08	Garage Update Process
BE-09	Parking Inventory Management

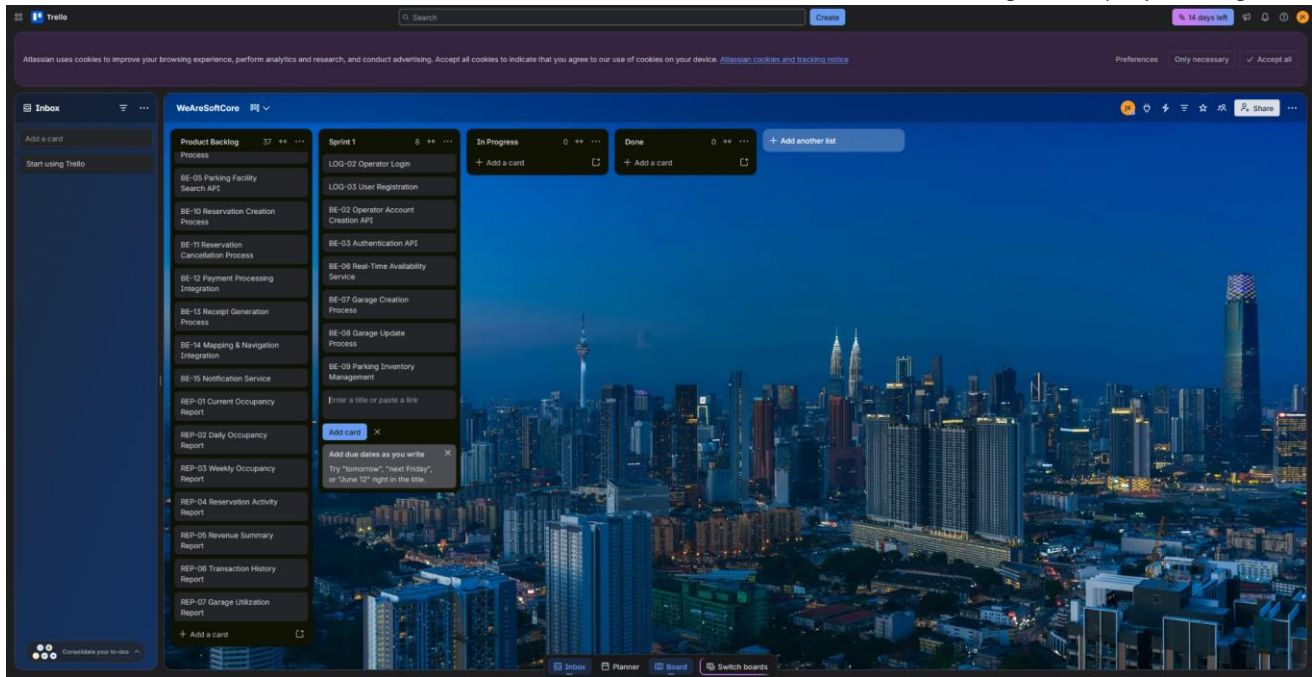


Figure 3. Trello board showing the Product Backlog and Sprint 1.